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Mobile Phone Use and Women's Empowerment in Bangladesh: An Analysis of the Bangladesh Financial Inclusion Insights Survey 2017

Ashim Kumar Nandi [1] and Ann-Zofie Duvander [2,*]

[1] Department of Sociology, University of Barishal, Barishal 8254, Bangladesh; aknandi@bu.ac.bd
[2] Department of Sociology, Stockholm University, S-106 91 Stockholm, Sweden
* Correspondence: ann-zofie.duvander@sociology.su.se

Abstract: Despite steady economic and social development in Bangladesh, women are in an underprivileged situation in many ways. This study explores the association between the use of information and communication technology (ICT) and women's empowerment in Bangladesh. This study employs ordinary least squares regression, the ordered logit model, and the multinomial logit model, using Financial Inclusion Insights (2017) Survey data, to investigate the relationship between mobile phone use and women's empowerment. The study's main result indicates that mobile phone use facilitates women's empowerment in general, but the impact needs to be considered for different groups of women. Housewives who are restricted within the household are impacted less than non-housewives by mobile phone use, contrary to our expectations. Also heads of households and spouses of heads of households, who are in very different positions in a patriarchal family structure, are similarly impacted by mobile phone use. This result implies that ICT only affects the empowerment of women who are not restricted to the household, and we conclude that ICT access alone is not enough to ensure women's awareness about their rights.

Keywords: mobile phone use; women's empowerment; patriarchy; household decision making; Bangladesh

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Academic Editor: Barbara Fawcett

Received: 13 December 2024
Revised: 15 February 2025
Accepted: 19 February 2025
Published: 25 February 2025

Citation: Nandi, Ashim Kumar, and Ann-Zofie Duvander. 2025. Mobile Phone Use and Women's Empowerment in Bangladesh: An Analysis of the Bangladesh Financial Inclusion Insights Survey 2017. *Social Sciences* 14: 139. https://doi.org/10.3390/socsci14030139

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1. Introduction

While a patriarchal family system may oppress women, feminist scholars argue that transforming women's awareness facilitates the realization of their potential and power and can help them resist oppression (Chasteen 2001; Whitehead 2007). Some scholars argue that mobile phones can help women realize their value to others (Chew et al. 2015) and redefine traditional gender roles (Garrido and Roman 2006; O'Neil and Cummings 2015). Some studies have also explored how mobile phones enhance women's empowerment (e.g., Hoan et al. 2016; Antonio and Tuffley 2014). Nevertheless, only some studies have examined the diverse effects of ICT on different groups of women or have questioned which women gain from mobile phone use (Chan 2015; Jeni et al. 2014).

This study explores whether mobile phone use is associated with women's empowerment. The key research questions in this paper are as follows: (1) Is mobile phone use associated with the empowerment of restricted women? (2) Is mobile phones associated with women's empowerment in male-headed households?

We employ indicators of empowerment in this study, such as household decision making and the ideas from Kabeer (1999) about operationalizing 'empowerment'. According to

(Kabeer 1999, p. 437), empowerment requires transformation. People with many options may be influential but not empowered, because they have never been powerless. Since women are in disempowered situations, this study focuses on the possibilities of women to be empowered.

We use Bangladesh as our case study as the use of mobile phones, one type of ICT, is becoming increasingly popular among its citizens (e.g., World Bank 2018). Almost two decades ago, the UN stated two essential facts regarding Bangladesh: first, gender-based restrictions confine women within the family, and, second, information and communication technologies (ICTs) can empower these women (United Nations 2005). However, how important ICT is and whether it impacts all groups of women in various different contexts remains to be found out.

Therefore, we ask, can ICT facilitate Bangladeshi women, especially those confined within the family and who live within a patriarchal family structure, to connect with other women, to become aware of their value and strengths, to resist oppression by men and, thereby, be empowered? The results also apply to other contexts involving a patriarchal social structure. The circumstances according to which ICT facilitates empowerment are crucial in many contexts.

Bangladesh is an exciting context for our investigation for various reasons. Firstly, there is a very high level of gender inequality in Bangladesh. The 2022 Gender Inequality Index ranked Bangladesh 127th out of 193 countries (United Nations Development Programme 2025).

Secondly, a gender gap persists, despite women's labor force participation (LFP) in Bangladesh climbing from 27.8% (85.5% for men) in 2005 (United Nations 2020) to 39.2% (81.4% for men) in 2022 (United Nations Development Programme 2025). Gender disparities persist despite some recent social and economic advances. While the LFP rate is higher for men (80.5%) than women (36.3%), the percentage of women (92%) in informal employment is higher than that of men (82%) (Bangladesh Bureau of Statistics 2022). Additionally, the average monthly income from employment (wage/salary) is higher among men than women (Bangladesh Bureau of Statistics 2022).

Thirdly, Bangladesh is experiencing fast growth in terms of mobile phone and internet use and there is a gender gap in this regard. According to the Population and Housing Census 2022 of Bangladesh, 69.93% of Bangladesh's total population (who are 15 years old and above) have mobile phones for their own personal use. Among the total male population (who are 15 years old and above), 84.16% have mobile phones for their own personal use, while 56.43% of females out of the total female population (who are 15 years old and above) have mobile phones for their own personal use (Bangladesh Bureau of Statistics 2023). The census also reported that 36.92% of Bangladesh's total population (who are 15 years old and above) use the internet. The census reported that 46.58% of the total male population (who are 15 years old and above) use the internet, while 27.77% of the total female population (who are 15 years old and above) use the internet (Bangladesh Bureau of Statistics 2023). In addition, between 2020 and 2021, mobile internet use among men in South Asia experienced continuous growth, while adoption among women stabilized, which resulted in an increase in the mobile internet gender gap in South Asia from 36% to 41% (Shanahan 2022).

Fourthly, Bangladesh's cultural, religious, and patriarchal family norms restrict women from leaving home on their own without a male accompanying them (Hossain and Beresford 2012). Research carried out in South Asian Islamic countries, including Bangladesh and Pakistan, reveals that women's experiences of technology are influenced by patriarchal practices, social norms, and religious beliefs, leading to negative experiences like cyber harassment and privacy concerns (Ahmed et al. 2022). In 2017, women headed

14.2% of Bangladeshi households, while men headed 85.8% (Bangladesh Bureau of Statistics 2019). Despite Bangladesh's economic and social progress, women are continuously found in a precarious situation.

Finally, the National Information and Communications Technology (ICT) Policy 2015 in Bangladesh is a crucial legal framework for 'Digital Bangladesh' initiatives. Through this vision, the Bangladesh government aimed to foster the growth of digital financial inclusion and women's empowerment with the help of information technology. The Financial Inclusion Insights Survey (FIIS) 2017, the survey data we analyzed in this paper, was motivated and conducted in regard to this context (e.g., InterMedia Bangladesh 2018). Evaluating the National ICT Policy 2015, Aziz (2020) concluded that the policy is ambiguous and techno-centric, and the context of the societal challenges in Bangladesh needs to be addressed for digital inclusion to be achieved. Analyzing the FIIS 2017 data to assess the effect of information technologies on women's empowerment is essential in this context.

2. Literature Review

2.1. The Link Between Mobile Phone Use and Women's Empowerment: Research Progress and Gaps

Mobile phone use may have diverse effects on women's empowerment. Summers et al. (2020), in a study of Maasai communities in Tanzania, found that phones have given some women power, and others have remained powerless, but neither is permanent. Some women have used phones to exercise their ongoing agency, while others have learned to use phones, only to have them taken away. Some women use phones to defy patriarchal conventions (e.g., flouting communication standards and creating enterprises), whereas others accept the status quo. Information and communication technologies (ICTs) can undermine patriarchal authority and promote women's empowerment, although this varies in regard to different contexts and individuals (Hussain and Amin 2018; Masika and Bailur 2015).

Existing studies have investigated the association between mobile phone use and women's empowerment (e.g., Rotondi et al. 2020; Dasuki et al. 2013). Access to smartphones has benefited women in terms of their personal and professional growth and has enhanced their knowledge of violence prevention in the country. Nevertheless, women's adherence to patriarchal norms, distrust of assistance organizations, and ongoing gender dynamics contribute to the constraints on technological integration in regard to its potential for addressing violence (Ahmed et al. 2022; Tasnima and Syed 2024). Rotondi et al. (2020) reported, analyzing data from 219 countries including advanced and developing countries and countries that respect gender equality and others that do not, that the spread of mobile phones is associated with women's empowerment. Smaller-scale studies also show that having a mobile phone has made it easier for women to obtain information about their sexual and reproductive health and has given them the power to make their own choices (Iacoella et al. 2022; Rotondi et al. 2020; O'Neil and Cummings 2015). In examining evidence from rural China, Zheng et al. (2022) also found that smartphones markedly enhance women's decision-making power, partly facilitated by the rise in off-farm jobs being carried out by women. Jayachandran (2015) contended that one indicator of decision-making authority is the self-reported data from female respondents to the DHS regarding their influence on household decisions concerning significant purchases. In less affluent countries, women are less likely to impact spending decisions. The income gradient observed among nations is also present inside nations. The DHS calculates a household wealth index tailored to each country. Women with wealth beyond the median for their country possess greater decision-making authority, and exhibit reduced tolerance for gender-based violence compared to those with income below the median. Ongo Nkoa

et al. (2022), based on evidence from Africa from 1996 to 2017, reported that the proliferation of ICT, as measured by mobile phone and internet penetration, substantially enhances women's political empowerment. Moreover, mobile phone use enhances women's empowerment, creating opportunities in regard to their health, education, and livelihood (Rahman et al. 2023; Mackey and Petrucka 2021; Samosir et al. 2020; Islam and Slack 2016).

The demographic and health survey (NIPORT and ICF 2024) uses women's attitudes toward wife-beating questions to assess women's empowerment. A recent study (Nandi et al. 2024) analyzing Bangladesh Demographic and Health Survey 2017–2018 data reported that household ownership of mobile phones is not independently associated with women's approval of wife-beating justifications, which suggests mobile phones are not facilitating women's empowerment in Bangladesh. However, another study (Al Juboori 2024), which analyzed the responses from the 2018 Iraq Multiple Indicator Cluster Survey, reported that owning a mobile phone was associated with a lower level of acceptance of domestic violence. These results indicate that the impact of mobile phone use on women's empowerment varies according to the context.

Mobile phone use may facilitate women's empowerment in various ways, for example, by expanding their social circles and networks for social support; by providing more information, especially on health and education services; by assisting in overcoming spatial, physical, and temporal boundaries (Rajkhowa and Qaim 2022); by protecting them from intimate partner violence (IPV); and by enhancing women's freedom of choice and action (Pesando 2022[1]; Hossain and Samad 2020[2]; Handapangoda and Kumara 2013[3]; Abraham 2009[4]; Hossain and Beresford 2012[5]; Macueve et al. 2009[6]; Bayes 2001[7]). However, there are various limitations in terms of the generalizability of these results. For instance, Handapangoda and Kumara's (2013) study only looked at specific groups of women, so the results might, for instance, not apply to wealthy homemakers. Although Pesando (2022) found that mobile phone use empowers women by protecting them from IPV (by increasing their decision-making power), the study does not reveal whether the effect of using phones varies according to different groups of women (e.g., women restricted within the household vs. others). However, the study acknowledged the importance of future research in countries that are culturally conservative and that have a patriarchal family structure, such as Bangladesh.

Bayes (2001) argues that women's status increased as a result of mobile phone use, because people were used to going to a female Phone Bari (house of the phone), where businesswomen were in charge of mobile phones. These visits were essential because only a few people had mobile phones 20 years ago. While Bayes' (2001) explanation for how to improve the status of women was helpful 20 years ago, today, the explanations in this field quickly need updating. At the time, visiting a businesswoman to use a mobile phone was considered a symbol of respect for the woman. Today, most families have their own mobile phones, which makes the Phone Bari business-related explanation less relevant. Moreover, Kyomuhendo (2009)[8] points out that mobile phone use gave women freedom from the inhibitions and structures imposed on them by the prevailing patriarchy. However, in Kyomuhendo's study, the role of mobile pay houses in women's empowerment makes it impossible the distinguish the role of mobile phones from business participation in empowering women.

Hossain and Beresford (2012) argue that women who use mobile phones have more control over technology and that, in the long run, this changes social, cultural, and religious norms. However, the same authors only examined one group of women: subscribers to Grameen Bank's village phone program. This group is representative of only some mobile phone users in Bangladesh today. Thus, we should be cautious about using the results to make comprehensive statements about Bangladesh today.

Mobile phone use helps women resist structural constraints through three mechanisms: avoidance, accommodation, and collaboration (Pei and Chib 2020). In regard to avoidance practices, a woman can 'block' men or guardians who want to monitor her. Women can also 'delete' messages as another form of avoidance practice (Pei and Chib 2020, p. 10). Mobile phone use can, thus, empower women by helping them deal with patriarchal pressure (Pei and Chib 2020, p. 9). Women also use accommodation strategies when they need help to avoid direct interventions by men, which forces women to face up to patriarchal oppression and control. For example, in female workplaces, mobile phones play a crucial role in integrating women into management structures. Female group leaders use online WeChat groups to communicate messages, collect work status reports, and criticize those who fail to meet performance standards. These activities allow them to effectively communicate with managers and improve workplace conditions (Pei and Chib 2020). Moreover, women use collaboration mechanisms to resist structural constraints. The potential of mobile phones fosters social connections and collective power among individuals. There are two types of collaboration practices: sharing informational and social support to promote women's social independence and challenging structural constraints as a manifestation of accumulated feminist power (Pei and Chib 2020, pp. 11–12).

Indicator 5.b.1 as part of the UN's Sustainable Development Goals (SDGs), which concerns "the proportion of individuals who own a mobile telephone, by sex," implies that increasing mobile phone use goes hand in hand with women's empowerment (United Nations 2025), but it does not explain why. Understanding what sustainable development means before assessing the progress made in regard to the SDGs is essential. While the SDGs aim to clarify sustainable development, the definition remains ambiguous and illusive (Freistein and Mahlert 2016). A few years after the Brundtland Report defined sustainable development and popularized the notion (World Commission 1987), Dobson (1996) found around 300 definitions and interpretations of the concept (Dobson 1996). To address this issue, Manioudis and Meramveliotakis (2022) proposed an analytical framework to analyze sustainable development, urging scholars to return to the legacy of the classical political economy. They suggest that this framework, based on the stage theories of Smith, Mill, and Marx, can contextualize and integrate various contemporary development studies and emphasizes the importance of history, interdisciplinary approaches, and social class analysis (Manioudis and Meramveliotakis 2022). Therefore, it is essential to understand whether the association between mobile phone use and women's empowerment varies according to women's backgrounds.

Moreover, mobile phone use can also disempower women by reinforcing gender-specific social and structural restraints that censor "women's agency through access, surveillance, and intervention" (Pei and Chib 2020, p. 8). To illustrate the surveillance mechanism, Pei and Chib (2020) exemplified that voice calls help men carry out confidential supervision of women. In addition, husbands can impose rules on their wives through the use of mobile phones when they are not physically in the same place. Another example is that fathers can monitor whether their daughters are dating secretly through the use of voice and video calls (Pei and Chib 2020).

The preceding discussion offers insight into the relationship between mobile phone use and women's empowerment, but also shows that there is no coherent evidence on how mobile phone use relates to women's empowerment, nor in regard to which groups of women are impacted. Therefore, more research is needed to determine how mobile phone use is associated with women's empowerment among different groups of women.

2.2. Hypotheses

We developed three hypotheses based on previous research that is relevant to the Bangladesh context. As discussed above, the association between women's mobile phone use and empowerment is important to assess in Bangladesh because, despite steady social and economic development, Bangladesh has a gender gap in regard to mobile phone use. Moreover, Hossain and Beresford (2012) claim that patriarchal family norms, as well as religious and cultural traditions, stop women from leaving home on their own. This context encouraged us to examine the abovementioned association by comparing various women's groups, such as housewives and non-housewives.

Earlier research has shown that decision-making indicators can measure empowerment (e.g., Tareque et al. 2011). Hence, we developed our hypotheses using decision-making indicators. Using the Kaiser criteria (please see the Section 3 for details), we selected an appropriate decision-making indicator from various similar indicators available in the dataset used in this study.

First, considering that Bangladeshi socio-cultural norms confine women within the family, it is reasonable to assume that using mobile phones will make it easier for women to participate in economic decision making within their households. Thus, *female mobile phone users are more involved in intra-household economic decision making than non-users (hypothesis 1).*

Second, we anticipate that the impact of mobile phone use will be more significant for housewives than working women, because housewives are more likely to be confined within the family. So, when housewives use mobile phones, it may affect their decision-making power more than non-housewives. Thus, *the difference in the probability of being involved in intra-household economic decision making between mobile phone users and non-users is higher for housewives than non-housewives (hypothesis 2).*

Third, generally, female heads of households are more involved in household decision making. So, mobile phone use is not likely to bring significant change in regard to the household head's involvement in household decision making. In contrast, the spouses of male heads of households are likely to be more dependent on their husbands, especially in regard to decision-making issues, due to a lack of knowledge about their rights. Since mobile phone use can give these women access to more information, the spouses of male heads of households will likely benefit more from mobile phone use than female heads of households. Thus, *the difference in the probability of being involved in intra-household economic decision making between female mobile phone users and non-users is higher for spouses of heads of households than female heads of households (hypothesis 3).*

3. Materials and Methods

3.1. Data Source

This study used the fifth wave of the Bangladesh Financial Inclusion Insights Survey 2017 (FIIS 2017), which collected data in July–September 2017. The Financial Inclusion Insights program is managed by InterMedia and funded by the Bill and Melinda Gates Foundation. The Bill and Melinda Gates Foundation owns all the data and materials generated by the initiative (InterMedia Bangladesh 2018). The data can be considered somewhat dated given that this is a subject where the situation changes constantly. However, the timing of the survey is ideal, in that it provides an overview of the situation because it occurred in the middle of several major changes. Bangladesh is, at this time, still a traditional and patriarchal society, and while mobile phone use has been quickly increasing, differences between such use according to gender and other characteristics are still occurring. It is, thus, an ideal time to measure how a new technology may impact various groups of women differently. The first author analyzed the data for his thesis in 2021, which we used for this paper in a modified form.

Although the Bangladesh Financial Inclusion Insights Survey 2018 (FIIS 2018) is the latest survey, the difference in the data collection time between FIIS 2017 (July–September 2017) and FIIS 2018 (September–October 2018) was a year, which is a very short time in which to capture any trends or changes in terms of science. Despite the abovementioned justification for using the FIIS 2017 data, we analyzed the FIIS 2018 data to confirm whether the new data have brought about any changes in regard to the acceptance/rejection decisions concerning the paper's hypotheses. The FIIS 2018 sample was constructed similarly to the 2017 survey, with the aim of generating dependable estimates for the primary indicators for the nation overall and for urban and rural regions individually (InterMedia 2019). Of the 6000 men and women interviewed, aged 15 and over and residing in households, the responses of 3466 women participants were analyzed. Place of residence, age group, marital status, education, and religion were the control variables. We employed the chi-square test to find associations between the control and outcome variables. All the variables in the final model (i.e., multinomial logit model) were variables according to which, when they were excluded, the change in the deviance compared with the corresponding χ^2 test statistic on the relevant degrees of freedom was significant (at the $p < 0.05$ significance level). Since the outcome variable was ordinal, we first selected the ordered logit model. Employing the Brant test, we found that the parallel regression assumption had been violated. Therefore, we used a multinomial logistic regression model for the final analysis. Based on the results, we confirm that hypothesis 1 (i.e., female mobile phone users are more involved in intra-household economic decision making than non-users) is accepted (please see Figure A1 in Appendix A). We also confirm that hypothesis 2 (i.e., the difference in the probability of being involved in intra-household economic decision making between mobile phone users and non-users is higher for housewives than non-housewives) is rejected (please see Figure A2 in Appendix A) and confirm that hypothesis 3 (i.e., the difference in the probability of being involved in intra-household economic decision making between female mobile phone users and non-users is higher for spouses of male heads of households than female heads of households) is also rejected (please see Figure A3 in Appendix A). These results are similar to the results from the analysis of the FIIS 2017 data.

3.2. Sampling Design and Sample Size

The survey utilized a stratified multistage sampling method to ensure the sample represented the entire country. To infer the target population (aged 15+), sampling weights were used, which were based on 2016 population projections by gender, age, and urban–rural residence (InterMedia Bangladesh 2018). The survey used a list of urban centers (e.g., division headquarters, other municipal centers, non-municipal centers or "Thanas") and villages as sampling frames and used stratified multistage sampling. First, the survey stratified each division by the urban/rural area. Then, the process selected 333 primary sampling units, including all eight divisional headquarters. After that, 24 households were selected from each "other municipal centers" and 16 households from each non-municipal center and village. In Dhaka's divisional headquarters, the eight largest wards were selected, using 38 households from each ward. Finally, one adult household member was randomly selected from each selected household. Among these adults, 49 percent were female. During face-to-face interviews, tablet devices were used to collect the data (InterMedia Bangladesh 2018).

Of the 6000 residents of Bangladesh who were contacted to take part in an interview, 4839 adults (2472 males and 2367 females) were interviewed, resulting in a response rate of 80.7 percent. The survey assessed financial inclusion as the proportion of persons (15 years and older) identified as possessing at least one account in their name with an institution that provides a comprehensive range of financial services and that is subject to govern-

mental regulation (InterMedia Bangladesh 2018). Thus, the sample included people aged 15 and older with active digital stored-value accounts, such as a bank account or an account with a non-bank financial institution that has digital access [a payment card, online access, or a mobile phone application] and a mobile money account. However, the survey excluded some individuals from the sample, namely those possessing accounts with non-full-service institutions, such as credit-only microfinance institutions (MFIs), those lacking a personal full-service account or utilizing another person's account, or those utilizing services such as money guards, savings collectors, and digital recharge cards not linked to a bank or microfinance institution account (InterMedia Bangladesh 2018). This study analyzed the data of 2367 female respondents to obtain the results.

3.3. Dependent Variable

From three relevant decision-making variables (i.e., "Roughly how involved or uninvolved are you typically in deciding how to spend your household's income?"; "Roughly how involved or uninvolved are you typically in deciding how your household's income is spent on basic needs like food and clothing?"; and "Roughly how involved or uninvolved are you typically in deciding how your household's income is spent on other things beyond basic needs?"), the outcome variable used in this study was selected based on the Kaiser criteria, which had an eigenvalue (i.e., 2.7) higher than 1, the factor loading and uniqueness values were 0.9502 and 0.0971, respectively, and it explained 91% of the variance. In addition, the outcome of the orthogonal varimax rotation retained this same variable, which showed that the variance was 2.7, the rotated factor loading was 0.9502, and the uniqueness was 0.0971. The total number of observations was 4839 in regard to the factor analysis of the abovementioned three decision-making variables. This study only used the Kaiser criteria (i.e., it includes factors with eigenvalues equal or higher than 1, since earlier studies suggested using any one option from different options in regard to factor analysis (e.g., Pohlmann 2004)). The outcome variable refers to the question, "How involved or not involved are you in deciding how to spend your household's income?" This question had five possible answers: "very uninvolved" (7.1% of respondents), "somewhat uninvolved" (14.7%), "neither uninvolved nor involved" (7%), "somewhat involved" (39.4%), and "very involved" (31.8%). We interpreted the last two categories of involvement as being empowered.

3.4. Key Independent Variables

"Own a phone" was the main independent variable, which was used to examine the effect of using a mobile phone in the context of the research questions. The respondents were asked, "Do you personally own a mobile phone?" 'Yes' (68.5%) or 'No' (31.5%) were the options for answering the question.

The question on work was aimed at testing hypothesis 2. The respondents were asked, "In the past 12 months, what were you mainly doing for work?" Each female respondent answered the question using any one of the following 11 categories: (1) working full-time for a regular salary (3.4%); (2) working part-time for a regular salary (1.1%); (3) working occasionally, irregular pay (whenever the work is available) (1.3%); (4) working per season (e.g., only during the harvest season) (.8%); (5) self-employed, working for yourself (4.8%); (6) not working, but looking for a job (.4%); (7) housewife or stay-at-home husband doing household chores (77.6%); (8) full-time student (6.6%); (9) not working because of retirement (1.3%); (10) not working because of sickness, disability, etc., (1.4%); and (11) other (1.4%). We created a *non-housewife* dummy variable by combining the categories 1, 2, 3, 4, 5, and 6. Following sensitivity analysis, categories 8, 9, 10, and 11 were eliminated because

they did not fit the purpose of the analysis. Students were generally excluded because they do not typically participate in household decision making.

The "relation to household head" was aimed to test hypothesis 3. The respondents were asked, "What is your relationship to the household head?" and were given seven options[9] to answer the question, such as the head (the respondent is head of the household) (9.9%), spouse (70.6%), son/daughter (11.7%), father/mother (4.9%), sister/brother (1.7%), grandchildren (0.1%), or other relatives (1%). Our chosen model added all the categories together into the estimation.

3.5. Control Variables

The study used 'education', 'age groups', and 'residence type' as the control variables in the analyses. Table 1 presents the details of how these variables are categorized.

Table 1. Descriptive statistics of females for all variables in terms of number and percent.

	Female N (%)
Roughly how involved or uninvolved are you typically in deciding how to spend your household's income?	
Very uninvolved	169 (7.1)
Somewhat uninvolved	348 (14.7)
Neither uninvolved, nor involved	165 (7.0)
Somewhat involved	932 (39.4)
Very involved	753 (31.8)
Do you own a phone?	
Yes	1622 (68.5)
No	745 (31.5)
In the past 12 months, what have you been doing for work?	
Housewife	1837 (77.6)
Non-housewife[10]	530 (22.4)
What is your relationship with the household head?	
Head (the respondent is head of the household)	235 (9.9)
Spouse	1672 (70.6)
Others family members/relatives	460 (19.4)
Education	
No formal education	448 (18.9)
Primary education	737 (31.1)
Secondary education	918 (38.8)
Higher education	257 (10.9)
Other	7 (0.3)
Age groups	
15–24	527 (22.3)
25–34	810 (34.2)
35–44	536 (22.6)
45–54	300 (12.7)
55 and over	194 (8.2)
Residence type	
Urban	667 (28.2)
Rural	1700 (71.8)
N	2367 (48.9)

Table 1 reports on the descriptive statistics of the female respondents for the dependent, independent, and control variables in terms of the number and percent of the respondents. The table indicates that most women are more or less involved in deciding how to spend their household income. Table 1 also shows that 68.5% of women own a phone, while 31.8% of women do not own a phone, in our sample. Among the female respondents, 77.6% are housewives.

3.6. Statistical Analysis

Ordered Logit Model and Multinomial Logit Model

Since the outcome variable in this study is an ordinal variable, we selected to use an ordered logit model (OLM) for analyzing the effect of mobile phone use on women's empowerment. The OLM can capture different distances between categories, such as the distance between 'very uninvolved' and 'somewhat uninvolved' or between 'somewhat involved' and 'very involved'[11]. We tested whether the proportional odds assumption (POA) was violated to assess the goodness of fit of the OLM. The POA, which is implicit in an OLM, indicates that the effect of b is the same regardless of the different categories compared (e.g., the effect of b for 1 vs. 2, 3, 4, 5, and 1, 2, 3, 4 vs. 5 are the same) in the following model:

$$\ln\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = a + \boldsymbol{X_i b}$$

$$\downarrow$$

$$1\ vs.\ 2,\ 3,\ 4,\ 5: ln\left(\frac{P(y_i>1)}{P(y_i\leq 1)}\right) = -\tau_1 + \boldsymbol{X_i b}$$

$$1,\ 2\ vs.\ 3,\ 4,\ 5: ln\left(\frac{P(y_i>1)}{P(y_i\leq 1)}\right) = -\tau_2 + \boldsymbol{X_i b}$$

$$1,\ 2,\ 3\ vs.\ 4,\ 5: ln\left(\frac{P(y_i>1)}{P(y_i\leq 1)}\right) = -\tau_3 + \boldsymbol{X_i b}$$

$$1,\ 2,\ 3,\ 4\ vs.\ 5: ln\left(\frac{P(y_i>1)}{P(y_i\leq 1)}\right) = -\tau_4 + \boldsymbol{X_i b}$$

The Brant test, introduced by Brant (1990), carried out using Stata 18, was employed to test the POA. A significant omnibus test statistic in the Brant test indicates that the POA is violated. If POA is violated, we can use the a multinomial logit model (MNLM) as an alternative to assess the ordinal outcome variable (Long and Freese 2014, p. 385). The MNLM can be imagined as "simultaneously fitting binary logits for all comparisons among the alternatives" (Long and Freese 2014, p. 386). In general, we used the MNLM when the POA was violated, otherwise we used the OLM.

Interpretation of the OLM and MNLM outputs is complicated; therefore, we used predicted probabilities based on the OLM and MNLM.

This study tests each of three hypotheses using two steps. In the first step, this study uses ordinary least squares (OLS) regression as a form of sensitivity analysis to find out whether one category of an independent variable is significantly different from another category. In the second step, this study employs either the OLM or MNLM and calculates the predicted probabilities. Next, we assessed whether the predicted probability of being involved in household decision making is higher for female mobile phone users than non-users. Suppose the study finds that female mobile phone users are significantly different from non-users, and the predicted probability of being involved in household decision making is higher for female mobile phone users, in that case, the study will conclude that hypothesis 1 is confirmed.

This study employed the *Pseudo-R^2*, likelihood ratio (LR) chi-square test in terms of all the coefficients, and the *likelihood-ratio (LR) test* to test the model fit.

3.7. Strengths and Limitations

A significant strength of this study is that it used nationally representative survey data (i.e., FIIS 2017), which includes data from samples of financially independent individuals. The nature of the sample helps to investigate how information technologies facilitate women's empowerment at the household level, particularly regarding those who have achieved, at least partially, economic empowerment.

There are some limitations in regard to the data. Firstly, the study used FIIS 2017 data, which cannot show the discussed effects of mobile phone use over time. Therefore, we must acknowledge the possibility of reverse causality, namely that household decision making affects mobile phone ownership. However, the data should uphold the importance of our study. We are focusing on the potential of mobile phone technology for women's empowerment, where the ownership of a mobile phone radically differs from other empowerment indicators. Empowered women will be more likely to have access to mobile phones and other symbols of wealth, such as expensive ornaments and fashionable dresses. However, these symbols have a different impact on their lives. As an information communication technology, mobile phones facilitate information gathering and connections to a broader network. Thus, mobile phones may empower women further. We propose that mobile phone use can take women from a lower level to a higher level of empowerment, so even if a woman is empowered without having a phone, she will likely be more empowered after having one. It may be possible to show whether mobile phones or empowerment come first, but mobile phones will likely empower women further.

Secondly, the sample is based on respondents with active digital stored-value accounts: a mobile money account, full-service bank, or NBFI account that offers digital services. This means that the samples do not represent those mobile phone users and non-users who do not have such accounts. To know the approximate size of the excluded group, we can look at the World Bank's 2017 Global Findex database. The database defines account ownership as having an individual or jointly owned account either at a financial institution or through a mobile money provider. In Bangladesh, nearly 50% of adults have such an account. Around 30% of adults used their accounts to make or receive digital payments using a mobile phone or the internet in the past year. Around 10 million account owners in Bangladesh use cash or an over-the-counter service to send or receive domestic remittances. The database also shows that 3% of adults are unbanked (i.e., remain without an account at a financial institution or through a mobile money provider), of whom more than half own mobile phones (Demirguc-Kunt et al. 2018). In addition to that, regarding those respondents with an account, the sample is weighted according to the population composition (see above).

Thirdly, as most mobile phones have applications in English, women with limited English language knowledge may benefit less from mobile phone use. These women may experience less empowerment due to mobile phone use, but there are also ways to overcome/minimize language barriers (e.g., using Google Translate or Bengali keyboards, such as Ridmik/Avro keyboards), and we estimate that this data limitation is minor.

Fourthly, data speed connectivity or mobile internet speed may impact the usage and choice of mobile applications. However, we could not estimate these effects because the variables/information were unavailable in the FIIS 2017 dataset.

Fifthly, the cost of mobile data (e.g., cost/Gb) can limit mobile phone use, which we could not estimate in the analysis due to the lack of relevant variables/information in our

dataset. Moreover, the categorization of apps may impact women's empowerment, which we also could not be estimated due to the lack of relevant information in the dataset.

Finally, the FIIS 2017 data is older than is ideal in terms of science. In recent years, many critical global events have taken place that could have significantly influenced the results. However, the findings are still relevant and necessary because the latest FIIS was conducted in 2018, which is also in the distant past.

4. Results

Out of 6000 individuals, aged 15 and older, who were eligible to be interviewed, 4839 (2472 males and 2367 females) participated in the interviews, yielding a response rate of 80.7 percent. The findings in this study were derived from 2367 female participants. The outcome variable asked respondents, "How involved or not involved are you in deciding how to spend your household's income?" Among the 2367 female respondents, 753 (31.8%) answered 'extremely involved', 932 (39.4%) answered 'somewhat involved', 165 (7%) answered 'neither uninvolved nor involved', 348 (14.7%) answered 'somewhat uninvolved', and 169 (7.1%) answered 'very uninvolved'.

First, we tested whether female mobile phone users are more involved in intra-household economic decision making than non-users. In the first step, the results of the OLS regression (please see Table 2) found that female mobile phone users are significantly different from female non-users. The model controlled for education, age, and residence. In the second step, the study found (please see Figure 1) that the predicted probability of being involved in deciding how to spend your household income is higher for female mobile phone users than non-users. Furthermore, the predicted probability of being uninvolved in decision making related to household income is lower for female mobile phone users than non-users.

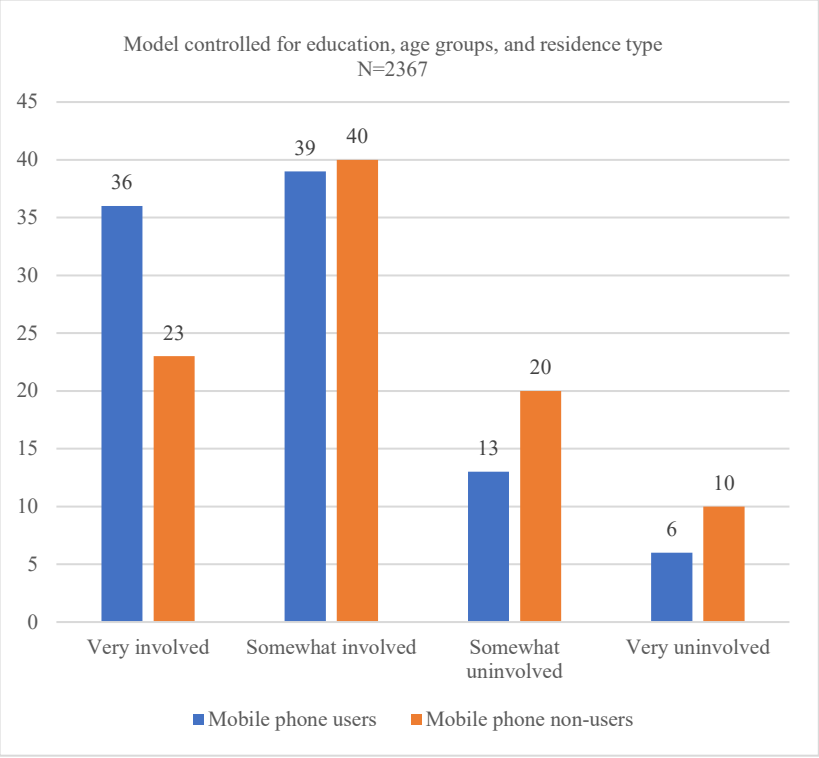


Figure 1. Predicted probability (percent) of how involved/uninvolved female mobile phone users and non-users are in deciding how to spend their household income, using ordered logit model.

Table 2. OLS regression of how involved/uninvolved females are in deciding how to spend their household income in terms of whether they own a mobile phone.

Independent Variable	Model 1 [Ψa]	Model 2 [bΨ]	Model 3 [aΨΨ]	Model 4 [bΨΨ]
Own a phone (No = ref)				
Yes	0.45 ***	0.41 ***	0.23 ***	0.21 ***
Constant	3.4 ***	2.9 ***	2.3 ***	2 ***
r^2	0.028	0.093	0.017	0.074
N	2367	2367	2367	2367

Note: [a] control variables not used in the measurement; [b] controlled for education, age groups, and residence type; *** $p < 0.001$; [Ψ] dependent variable (i.e., GN2_1) used original categories including 1 = very uninvolved, 2 = somewhat uninvolved, 3 = neither uninvolved nor involved, 4 = somewhat involved, 5 = very involved; [ΨΨ] dependent variable (i.e., GN2_1) is recoded as 1 = very uninvolved + somewhat uninvolved, 2 = neither uninvolved nor involved, and 3 = very involved + somewhat involved.

Since female mobile phone users are significantly different from non-users and the predicted probability of females being involved in deciding how to spend their household income is higher for female mobile phone users, this study concludes that hypothesis 1 is confirmed.

Second, we tested whether the difference in the probability of being involved in intra-household economic decision making between mobile phone users and non-users is higher for housewives than non-housewives. This study found, by employing OLS regression (please see Table 3), that mobile phone users who are housewives are significantly different from non-users, when controlling for education, age groups, and residence type. On the other hand, mobile phone users who are not housewives are significantly different from non-users who are not housewives in the model that controlled for education, age groups, and residence type.

Table 3. OLS regression of how involved/uninvolved mobile phones users and non-users who are housewives and non-housewives are in deciding how to spend their household income.

Independent Variables	Model 1 [Ψa]	Model 2 [bΨ]	Model 3 [aΨΨ]	Model 4 [bΨΨ]
Recoded variable (housewife non-user = ref)				
Housewife user	0.16 **	0.15 **	0.08 *	0.07 *
Constant	3.8 ***	3.3 ***	2.5 ***	2.3 ***
r^2	0.004	0.030	0.023	0.023
N	2116	2116	2116	2116
Recoded variable (non-housewife non-user = ref)				
Non-housewife user	0.31 ***	0.30 ***	0.11 *	0.11 *
Constant	3.8 ***	3.3 ***	2.6 ***	2.3 ***
r^2	0.006	0.032	0.002	0.023
N	2116	2116	2116	2116

Note: [a] control variables not used in the measurement; [b] controlled for education, age groups, and residence type; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; [Ψ] dependent variable (i.e., GN2_1) used original categories including 1 = very uninvolved, 2 = somewhat uninvolved, 3 = neither uninvolved nor involved, 4 = somewhat involved, 5 = very involved; [ΨΨ] dependent variable (i.e., GN2_1) is recoded as 1 = very uninvolved + somewhat uninvolved, 2 = neither uninvolved nor involved, and 3 = very involved + somewhat involved.

Figure 2 presents the finding that the difference in the predicted probability of being involved in deciding how to spend their household income between mobile phone users and non-users is lower for housewives than non-housewives, when controlling for education, age groups, and residence type. On the other hand, the difference in the predicted probability of being uninvolved in household decision making between mobile phone

users and non-users is slightly higher for housewives than non-housewives, when controlling for education, age, and residence.

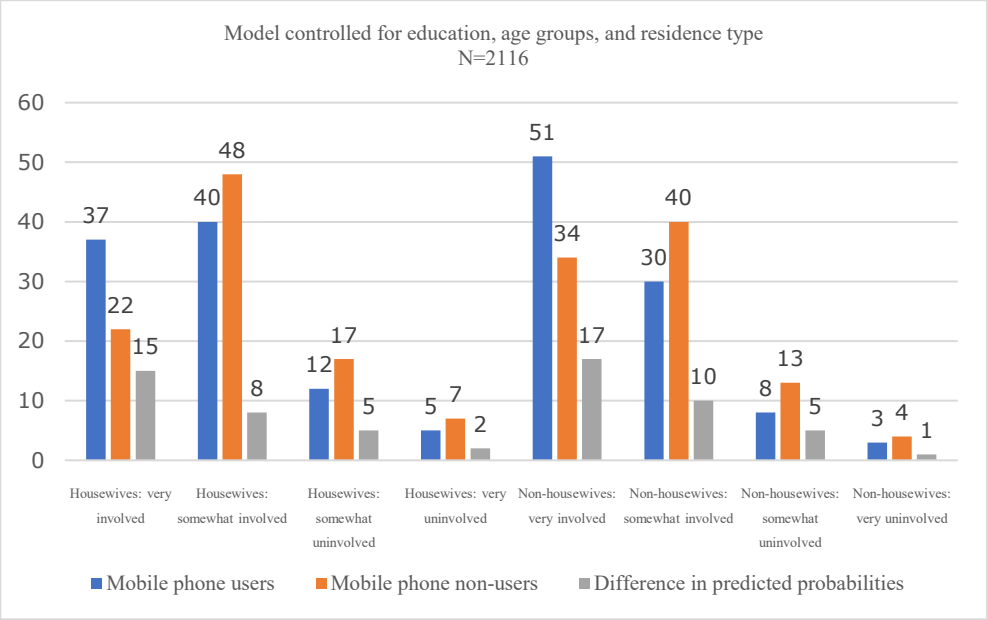


Figure 2. Difference in predicted probability (percent) of how involved/uninvolved mobile phone users and non-users who are housewives and non-housewives are in deciding how to spend their household income, using the multinomial logit model.

Although both mobile phone users who are housewives and non-housewives are significantly different from non-users, the difference in the predicted probability of being involved in deciding how to spend their household income between mobile phone users and non-users is lower for housewives than non-housewives. So, this study rejects hypothesis 2.

Lastly, we tested whether the difference in the probability of being involved in intra-household economic decision making between female mobile phone users and non-users is higher for spouses of heads of households than female heads of households. In the first step, the results of the OLS regression (please see Table 4) showed that female mobile phone users who are spouses to the head of the household are significantly different from females that do not use a mobile phone who are spouses to the head of the household, when controlling for education, age groups, and residence type. In addition, the study found that female mobile phone users who are the head of the household are significantly different from females that do not use a mobile phone who are the head of the household.

We found (please see Figure 3)[12] that the difference in the predicted probability of being involved in deciding how to spend their household income between female mobile phone users and non-users is lower for spouses of the head of the household than female heads of the household. On the other hand, the difference in the predicted probability of being uninvolved in the household decision-making between female mobile phone users and non-users is higher for spouses of the head of the household than female heads of the household. The model was controlled for education, age groups, and residence type.

Both female mobile phone users who are spouses of the household head and female household heads are significantly different from non-users. However, this study finds that the difference in the predicted probability of being involved in deciding how to spend their household income between female mobile phone users and non-users is lower for spouses of household heads than female heads of the household. Hence, this study rejects hypothesis 3.

Table 4. OLS regression of how involved/uninvolved female mobile phone users and non-users who are spouses of the head of the household and females who are the head of the household are in deciding how to spend their household income.

Independent Variables	Model 1 [Ψa]	Model 2 [bΨ]	Model 3 [aΨΨ]	Model 4 [bΨΨ]
Recoded variable (spouse non-user = ref)				
Spouse user	0.41 ***	0.30 ***	0.28 ***	0.21 ***
Constant	3.5 ***	3 ***	2.4 ***	2.1 ***
r^2	0.027	0.084	0.028	0.076
N	2367	2367	2367	2367
Recoded variable (head non-user = ref)				
Head user	0.83 ***	0.73 ***	0.33 ***	0.27 ***
Constant	3.7 ***	3.1 ***	2.5 ***	2.1
r^2	0.033	0.096	0.012	0.069
N	2367	2367	2367	2367

Note: [a] control variables not used in the measurement; [b] controlled for education, age groups, and residence type; *** $p < 0.001$; [Ψ] dependent variable (i.e., GN2_1) used original categories including 1 = very uninvolved, 2 = somewhat uninvolved, 3 = neither uninvolved nor involved, 4 = somewhat involved, 5 = very involved; [ΨΨ] dependent variable (i.e., GN2_1) is recoded as 1 = very uninvolved + somewhat uninvolved, 2 = neither uninvolved nor involved, and 3 = very involved + somewhat involved.

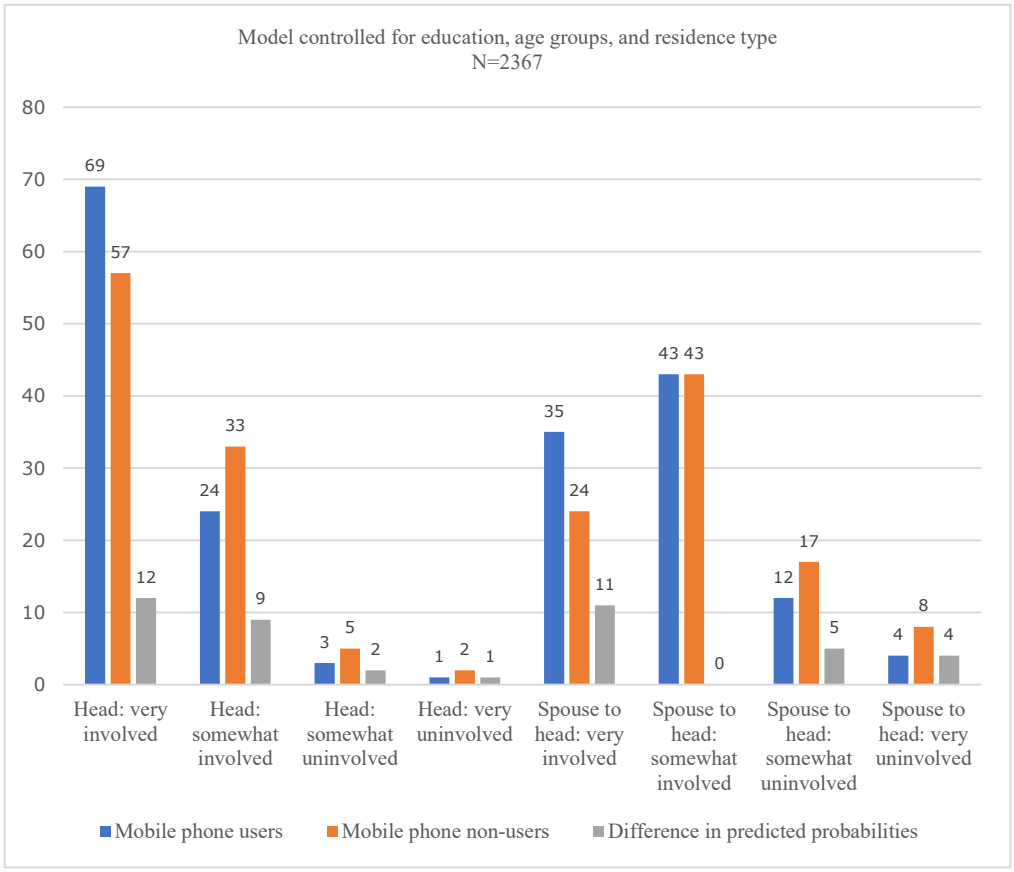


Figure 3. Difference in predicted probability (percent) of how involved/uninvolved female mobile phone users and non-users who are spouses of household heads and female heads of the household are in deciding how to spend their household income, using the ordered logit model.

5. Discussion

This study investigates the association between mobile phone use and women's empowerment. Women's empowerment, the dependent variable, was assessed using a question on how involved or uninvolved the respondents typically were in deciding how to spend their household income (the categories of answers were very uninvolved, somewhat uninvolved, neither uninvolved nor involved, somewhat involved, and very involved). Of the 2367 women in this study, 932 (39.4%) and 753 (31.8%) women were somewhat involved and very involved in deciding how to spend their household income, respectively. The main results in this study indicate that mobile phone use facilitates women's empowerment; mobile phone-using women are more involved in household decision making than women that do not use a mobile phone. We conclude that women are being empowered to some degree. This result was consistent with the findings in some earlier studies (e.g., Al Juboori 2024; Ongo Nkoa et al. 2022; Rotondi et al. 2020), while being dissimilar to the findings by Nandi et al. (2024). One plausible reason for the contradictions in the results in comparison with previous studies is that the empowering effects of information and communication technologies (ICTs) depend on which women and in which context women use the technology (O'Neil and Cummings 2015). However, this study finds a lack in variation in the association with mobile phone use for various groups. For example, mobile phone use does not provide additional benefits to disadvantaged groups, such as housewives, in terms of their empowerment. This result implies that there might be other constraints, such as conservative family norms, which could also play a crucial role in achieving women's empowerment that might not be overcome through the provision of information alone. Another interpretation of this result is that although Handapangoda and Kumara (2013) argued that mobile phone use assists women's empowerment by overcoming spatial, physical, and temporal boundaries, the impact is not more substantial for those women that are restricted within the household.

Moreover, this study finds that spouses within male-headed households do not receive more benefits than female heads of households when using mobile phones for their empowerment. Mobile phone use has no additional importance in regard to helping women's empowerment in male-headed households. This result suggests that although mobile phone use helps women's empowerment, it does not provide additional support to disempowered women subject to a patriarchal family structure. This finding is consistent with that of O'Neil and Cummings (2015), who reported that women's access to and utilization of digital ICTs may alter gender-based power dynamics. One interpretation of this result might be that excessive male dominance and women's belief in patriarchal norms can undermine the benefit of accessing information and can limit technological integration that could be used to respond to violence (Tasnima and Syed 2024).

Some earlier studies (e.g., Hossain and Samad 2020; Handapangoda and Kumara 2013; Hossain and Beresford 2012) claim that mobile phone use positively impacts women's empowerment, since it assists women to access information. However, the result in this study indicates that any information that women get via mobile phones does not enhance women's involvement in household decision making in situations where they are already disadvantaged, such as housewives and those with a male head of the household. Information accessed via mobile phones facilitates women's empowerment but does not help particular groups, such as restricted women, by changing their perception of their own rights within the family. So, mobile phone use does not equalize the situation of women in Bangladesh, but the differences between these groups of women remain. These findings may be generalizable to all patriarchal societies where women are confined within the household and where mobile phone use has been quickly increasing.

Some scholars (e.g., Hossain and Samad 2020; Handapangoda and Kumara, 2013; Bayes 2001) have argued that mobile phone use facilitates women's empowerment by enhancing women's freedom of choice and action. The results in this study show that mobile phone use essentially helps women's empowerment through helping them attain a higher level of involvement in household decision making. However, the results in this study should caution against assuming any general impact and that subgroups of women, such as housewives and those with a male head of the household, need to be studied separately.

We do not conclude that mobile phone use is unimportant for empowerment for all groups of women. If women can challenge norms through mobile phone-mediated interactions, mobile phone use can be important. The reason for such an argument is that socio-cultural norms restrict millions of women within the family. In addition, the patriarchal family structure oppresses women in terms of their rights. In this context, if mobile phone-mediated interaction helps women to challenge gendered norms through building awareness about their rights, mobile phone use can be an essential facilitator of women's empowerment. This study finds a blend of results in this regard. Although mobile phone use helps women's empowerment in general, the impact shows a lack of variation for various groups.

6. Conclusions

This paper investigates the association between mobile phone use and women's empowerment in Bangladesh. Women's involvement in deciding how to spend their household income was used as an indicator of women's empowerment. Of the 2367 women interviewed, 39.4% and 31.8% were somewhat involved and very involved in deciding how to spend their household income, respectively. The paper reports that 68.5% of the 2367 women personally own a mobile phone. The study tested three hypotheses to explore the association between mobile phone use and women's empowerment, controlling for potential confounding factors (i.e., age, education, and residence type). The first hypothesis, that female mobile phone users are more involved in intra-household economic decision making than non-users was accepted. The study rejected the second hypothesis, namely that the difference in the probability of being involved in intra-household economic decision making between mobile phone users and non-users is higher for housewives than non-housewives, and the third hypothesis, namely that the difference in the probability of being involved in intra-household economic decision making between female mobile phone users and non-user is higher for spouses of heads of the household than female household heads.

6.1. Implications for Theory, Policy, and Practice

Ensuring gender equality and women's empowerment is vital to Bangladesh's constitution and policy. Article 10 of the constitution of Bangladesh states that the participation of women in all spheres of national life should be ensured (Government of the Peoples' Republic of Bangladesh 2016). Moreover, some of the objectives in the National Women Development Policy 2011 of Bangladesh have been designed in regard to female empowerment "to ensure the socio-economic, political, administrative, and legal empowerment" (objective 16.3), "to ensure full and equal participation of women in the mainstream socio-economic development" (objective 16.5), "to remove existing male–female disparities" (objective 16.8), and "to eliminate discrimination in regard to women and female children" (objective 16.11). The policy also suggests preparing necessary legislation in favor of women's rights in the field of technology (Government of the Peoples' Republic of Bangladesh 2011). Moreover, achieving gender equality is the goal 5 of the Sustainable Development Goals (SDGs) of the United Nations, and target 8 of the goal is to "enhance the use of enabling

technology, in particularly information and communications technology, to promote the empowerment of women" (United Nations Development Programme 2024).

The current study contributes to the literature examining the association between mobile phone use and women's empowerment. The study results indicate that mobile phone use empowers women, but not more so for disadvantaged groups (e.g., housewives and spouses in male-headed households) than others. The inequality between women remains. Moreover, the results imply that although mobile phone use enhances women's decision-making power, it may not transform women's awareness, which can help them resist oppression generated by the patriarchal family system, which is inconsistent with existing theoretical views (e.g., Chasteen 2001; Whitehead 2007; Garrido and Roman 2006; O'Neil and Cummings 2015; Pei and Chib 2020).

Previous research has found that business training using online applications was positively associated with vulnerable groups' (e.g., migrants) personal growth and empowerment through digital literacy, professional skills, and vocational training (Drydakis 2022; Drydakis 2021; UNESCO 2018; Marchand and Dijkhuizen 2018; Sarvimäki and Hämäläinen 2016), and policymakers may consider offering empowering or awareness app training to vulnerable groups, such as housewives or spouses in male-headed households, to enhance the effects of mobile phone use on their empowerment. Moreover, the results in this study imply that enhancing the ownership of mobile phones among women in Bangladesh and countries with similar contexts will facilitate their empowerment. Acquiring digital skills and utilizing mobile phones when needed is essential, and such technology should be circulated among women.

6.2. Implications for Future Research

Future studies should continue to investigate the effects of mobile phone use and other new technologies on women's disempowerment. Moreover, as we conclude that mobile phone use confers no additional benefits to housewives and females in male-headed households, additional research can be conducted to determine what other factors, such as traditional family norms or excessive male dominance or religious views, mitigate the positive effects of mobile phone use. We also suggest that future research should explore how women use mobile phones, how women experience mobile phone use, and the danger of being surveilled through the same means.

Author Contributions: Conceptualization, A.K.N.; methodology, A.K.N.; software, A.K.N.; validation, A.K.N.; formal analysis, A.K.N.; investigation, A.K.N.; resources, A.K.N.; data curation, A.K.N.; writing—original draft preparation, A.K.N.; writing—review and editing, A.-Z.D.; visualization, A.K.N.; supervision, A.-Z.D.; project administration, A.K.N.; funding acquisition, A.K.N. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by [the Swedish Institute] grant number [13910/2019]. The authors received no external funding for APC.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data associated with this study is available at https://www.finclusion.org/data_fiinder.html (accessed on 1 February 2025).

Conflicts of Interest: The authors declare no conflict of interest.

Appendix A. Results Based on the Bangladesh Financial Inclusion Insights Survey 2018 Analysis

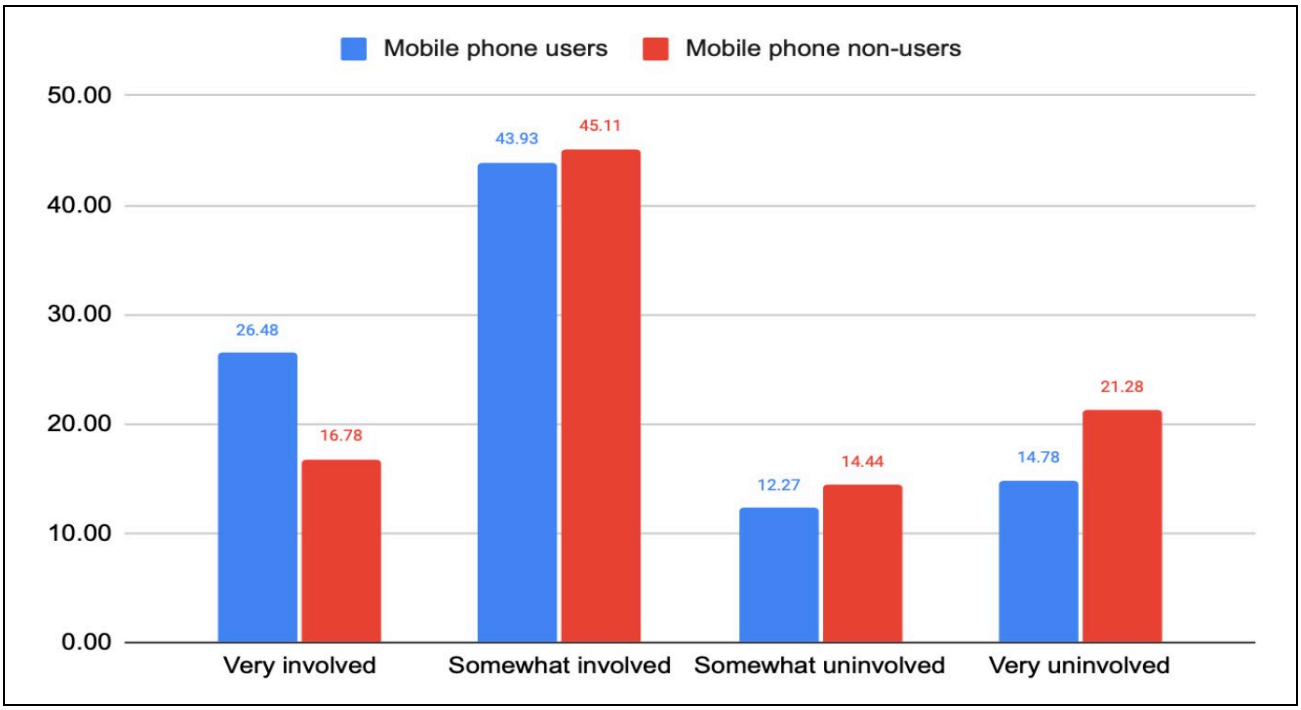


Figure A1. Predicted probability (percent) of *how involved/uninvolved female mobile phone users and non-users are in deciding how to spend their household income*, using the multinomial logit model (n = 3330). Adjusted for relationship to household head, working status, place of residence, age group, marital status, education, and religion. All variables in the multinomial logit model were variables for which, when excluded, the change in deviance compared with the corresponding χ^2 test statistic on the relevant degrees of freedom was significant.

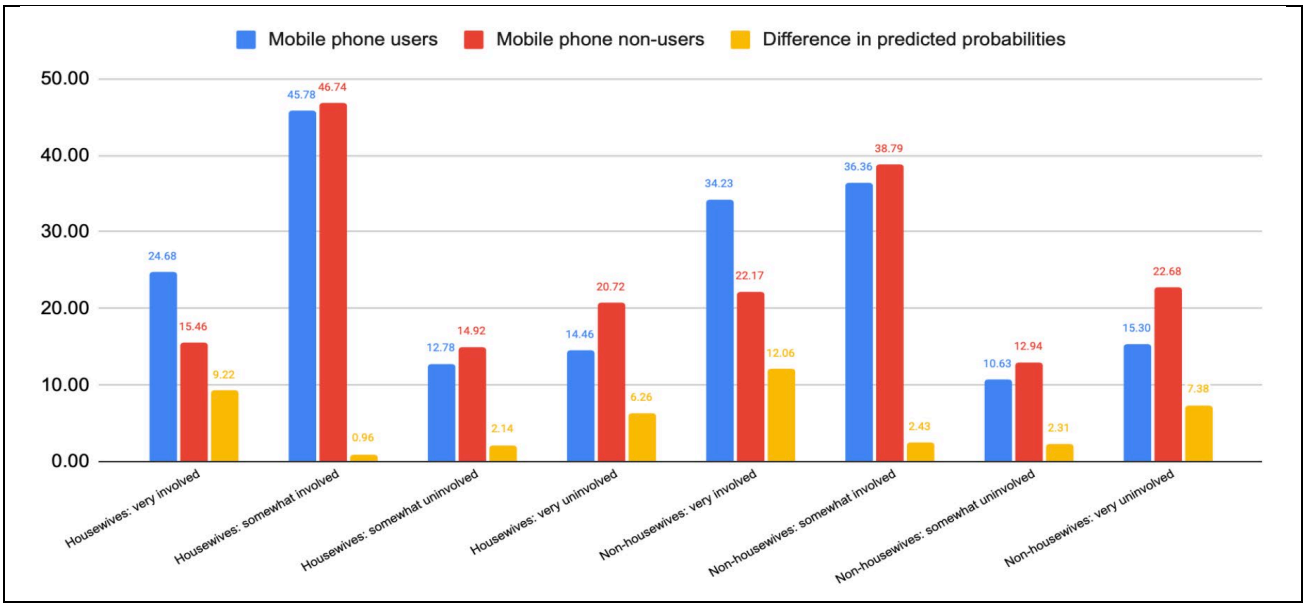


Figure A2. Difference in predicted probability (percent) of *how involved/uninvolved female mobile phone users and non-users who are housewives and non-housewives are in deciding how to spend their household income*, using multinomial logit model (n = 3330). Adjusted for place of residence, age group, marital

status, education, and religion. All variables in the multinomial logit model were variables for which, when excluded, the change in deviance compared with the corresponding χ^2 test statistic on the relevant degrees of freedom was significant.

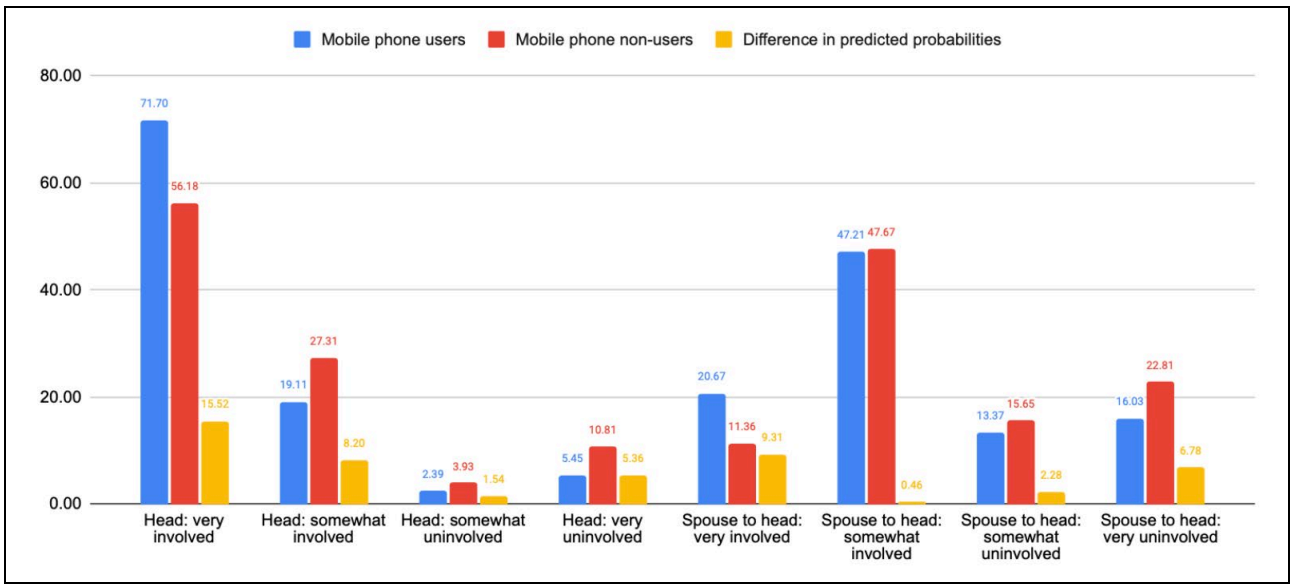


Figure A3. Difference in predicted probability (percent) of *how involved/uninvolved female mobile phone users and non-users who are spouses of household heads and female household heads are in deciding how to spend their household income*, using the multinomial logit model (n = 3330). Adjusted for place of residence, age group, marital status, education, and religion. All variables in the multinomial logit model were variables for which, when excluded, the change in deviance compared with the corresponding χ^2 test statistic on the relevant degrees of freedom was significant.

Notes

1 This study was conducted to examine the association between mobile phone use and intimate partner violence using demographic and health survey data from 10 low- and middle-income countries.

2 This study was conducted in remote rural areas of Bangladesh to examine the effect of mobile phone use on household welfare and women's empowerment.

3 This study was conducted involving poor housewives in Sri Lanka.

4 This study was conducted to examine the role of mobile phones in advocating women's rights in Zambia.

5 This paper was a reviewed paper that was designed to address the potential of ICT to emphasize Grameen Bank's village phone programme in achieving women's empowerment in Bangladesh.

6 This study was conducted in Mozambique.

7 This study was conducted to examine Grameen Bank's village phone initiative in Bangladesh.

8 This study was conducted to analyze the role of mobile payphone businesses on women's empowerment in Uganda.

9 The percentage distribution, which is presented within the brackets for each category, are calculated using the data of the female respondents.

10 In the table, the *non-housewives* dummy variable included all categories except 7, i.e., categories 1, 2, 3, 4, 5, 6, 8, 9, 10, and 11. After excluding categories 8, 9, 10, and 11, the descriptive statistics are *non-housewives* = 279 (13.2%) and *housewives* = 1837(86.8%), which are not presented here but can be provided by the authors.

11 Please note that (Winship and Mare 1984, pp. 521–523) showed that the linear regression model (LPM) is not an appropriate model for an ordinal outcome.

12 Please note that although Model 1 violated the parallel regression assumption, the study used OLM regression for measuring the model because Model 2 did not violate the assumption. So, to keep the results comparable, the study used OLM regression for both models. Moreover, the study measured Model 1 using MNLM regression and found similar results as the OLM regression.

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